

A developer wants to use an algorithm that uses a tree for their machine learning task. He has five classes to classify. Which of the following algorithm does he use?

- ☐ Decision tree
- ☐ SVM
- ☐ B-tree
- ☐ There is no algorithm which uses tree

Emily wants to create a n-D array with dtype=int32. Which of the following code is correct for this purpose?

- ☐ np.array([list1, list2])
- ☐ np.array([list1, list2], np.int32)
- ☐ np.array([list1, list2], np.int)
- ☐ np.array([list1, list2])

A developer wants to calculate the Cumulative Distribution Function of 'point 5'. What is the correct option to be used here?

- ☐ from scipy.stats import norm
print(norm.cdf(5))
- ☐ from scipy.stats import cdf
print(norm.cdf(5))
- ☐ from scipy.stats import cdf
print(cdf(5))
- ☐ from scipy.stats import norm
print(norm.Cdf(5))

Ronaldo wants to develop a system capable of recognizing two people's faces. Which of the following machine learning model can he use to achieve this functionality?

- ☐ Regression
- ☐ Neural network
- ☐ Clustering
- ☐ KNN

Which of the following tools are used in NLP to perform operations on natural language (English) like sentence structuring, finding root words, etc.?

- ☒ NLTK
- ☐ Text extraction
- ☐ Machine translation
- ☐ There is no tool for this
- Clear Response

A machine learning engineer has a dataset with four columns - flat no., area, price, property. He wants to change the column names from flat no. and price with house and value respectively.

Which of the following code is correct for this purpose?

- ☐ df = pd.read_csv('data.csv')
df = df.rename(columns={"flatno": "house", "price": "value"})
- ☐ df = pd.read_csv('data.csv')
df = df.rename(column={"flatno": "house", "price": "value"})
- ☐ df = pd.read_csv('data.csv')
df = df.rename(column={"flatno": "house", "column": "price": "value"})
- ☐ df = pd.read_csv('data.csv')
df = df.chrename(column={"flatno": "house", "column": "price": "value"})

Ronaldo wants to develop a system capable of recognizing two people's faces. Which of the following machine learning model can he use to achieve this functionality?

- ☐ Regression
- ☐ Neural network
- ☐ Clustering
- ☐ KNN

A developer has a dataset with three columns - age, dob, and profession - in which the age column has a missing value. A developer wants to fill this missing value by providing the mean of age.

Which of the following code does he use?

- ☐ df = pd.read_csv('data.csv')
df.age = df.age.fillna(np.mean(df.age))
- ☐ df = pd.read_csv('data.csv')
df.age = df.age.fill(np.mean(df.age))
- ☐ df = pd.read_csv('data.csv')
df.age = df.fillna(np.mean(df.age))
- ☐ df = pd.read_csv('data.csv')
df.age = np.mean(df.age)

Cobalt has a dataset with columns name, age, dob, and salary. While observing the data set, he observed many missing values in the column salary.

Which of the following code shows us the number of missing value in the salary column?

- ☐ sum(df['salary'].isnull())
- ☐ (df['salary'].isnull())
- ☐ sum(df['salary'].null())
- ☐ avg(df['salary'].isnull())



Join our channel if you haven't joined yet https://t.me/fresco_milestone (@fresco_milestone)

Cobalt has a dataset with columns name, age, dob, and salary. While observing the data set, he observed many missing values in the column salary.

Which of the following code shows us the number of missing value in the salary column?

- ☐ `sum(df['salary'].isnull())`
- ☐ `(df['salary'].isnull())`
- ☐ `sum(df['salary'].null())`
- ☐ `avg(df['salary'].isnull())`

Rick creates the numpy array and stores it in variable n as follows:

```
array_A = np.array([ [1,2,3], [1,2,3] ])
```

He wants to know the shape of the above array. Which of the following code should he use?

- ☐ `n.shape()`
- ☐ `n.shape`
- ☐ `n.size()`
- ☐ `n.size`

Ryan is working on a dataset and notices that the data is scaled differently. He tries to rescale the attributes with a mean of 0 and a variance of 1. What method did he apply, and what package did he implement?

- ☐ Standardization, SciPy
- ☐ Normalization, SciPy
- ☐ Standardization, Scikit-learn
- ☐ Normalization, Scikit-learn

Jessica is working on a Numpy array and she wants to perform some mathematical operations on it. For this, she writes the following code:

```
import numpy as np
a = np.array([100,200,300,400,500])
b = np.array([3,4,5,6])
print(np.subtract(a, b))
```

What will be the output of the above code?

- ☐ `[ 97 196 295 394 500]`
- ☐ Value error
- ☐ Invalid operation
- ☐ `[100,200,300,400,500]`

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

A developer is working on data preprocessing and wants to scale his dataset between a given minimum and maximum value. Which of the following scikit-learn modules should he use for this?

- ☐ `sklearn.preprocessing.StandardScaler()`
- ☐ `sklearn.preprocessing.MinMaxScaler()`
- ☐ `sklearn.preprocessing.QuantileTransformer()`
- ☐ There is no such module

Analyze regular expressions for the following:

Any binary string, e.g. Strings like the below should be accepted:

```
1
10
110001011001
101010101
```

While strings like the below should be rejected:

```
0
00
Dog 153
Hello
Chennai26
```

- ☐ `(1+0)*`
- ☐ `111 (1+0)*`
- ☐ `1 ((1+0)*1)*`
- ☐ `1 (1+0)*`

David has a dataset with six columns id, age, salary, designation, company, and profession. While working on this dataset, he observed that there were some columns like id and company that did not have any usual information, so he wanted to drop them. He stored this dataset into the data frame variable df. Which of the following codes should he use for this purpose?

- ☐ `df.drop(['id','company'], axis=1)`
- ☐ `df.drop(['id','company'], axis=-1)`
- ☐ `df.dropna(['id','company'], axis=1)`
- ☐ `df.drop('id','company', axis=1)`



A survey on the average amount of time spent on smartphones was conducted. Typically, on the left, you see a list of all the things or activities people do with their phones, followed by a graph showing how much time they spend doing them. Which type of chart should be used for this purpose?

- ☐ `matplotlib.pyplot.hist()`
- ☐ `plt.bar(x, height, width, bottom, align)`
- ☐ `matplotlib.pyplot.pie(data, explode=None, labels=None, c`
- ☐ `matplotlib.pyplot.scatter()`

While working on a movie review dataset, we take three reviews and perform a bag of words. What are the vectors of review1, review2, and review3?

review 1: This movie is very scary and king
review 2: This movie is not scary and is slow
review 3: This movie is spooky and good

- ☐ Vector of Review 1: [1 1 1 1 1 1 0 0 0 0]
Vector of Review 2: [1 1 2 0 0 1 1 0 1 0 0]
Vector of Review 3: [1 1 1 0 0 0 1 0 0 1 1]
- ☐ *Vector of Review 1: [1 0 1 0 0 0 1 0 0 1 0]
Vector of Review 2: [1 1 3 0 0 1 1 0 1 0 0]
Vector of Review 3: [1 1 0 1 0 1 1 1 0 0 0]
- ☐ Vector of Review 1: [1 1 1 1 1 1 1 0 0 0]
Vector of Review 2: [1 1 1 0 0 0 1 0 0 1 0]
Vector of Review 3: [1 1 2 0 0 1 1 0 1 0 0]
- ☐ Vector of Review 1: [1 0 1 0 0 0 1 0 0 1 0]
Vector of Review 2: [1 1 0 1 0 1 1 1 0 0 0]
Vector of Review 3: [1 1 3 0 0 1 1 0 1 0 0]

While working on a machine learning model, it is necessary to convert a 2-dimensional or multi-dimensional array into a contiguous flattened array for various calculations. For this, which NumPy operation will be used?

- ☐ `numpy.shape(a)`
- ☐ `numpy.ravel(a, order='C')`
- ☐ `numpy.rollaxis(a, axis, start=0)`
- ☐ `numpy.swapaxes(a, axis1, axis2)`

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

As a senior ML engineer, Rio is working on a cardiac prediction dataset with his juniors which he wants to share. When the dataset has been sent, the column names cause everyone to be puzzled. He decides to modify the names, but the syntax confuses him. Which is the correct syntax for this?

- ☐ `df.rename(columns = {'old_col1':'new_col1', 'old_col2':'new_col2'}, inplace = True)`
- ☐ `df.rename(columns = {'old_col1':'new_col1', 'old_col2':'new_col2'}, inplace = False)`
- ☐ `df.columns = df.columns.str.replace('old_char', 'new_char')`
- ☐ `df['Date'] = pd.to_datetime(df['Date'])`

Clark has an employee dataset in which four columns are present, namely first name, last name, age, and salary. He wants to set the first name as an index column and store his dataset into a variable emp. Which of the following codes should he use for this purpose?

- ☐ `emp.set_index("firstname", inplace = True)`
- ☐ `emp.set_index("firstname", inplace = False)`
- ☐ `emp.index="firstname"`
- ☐ There is no such code



Join our channel if you haven't joined yet https://t.me/fresco_milestone (@fresco_milestone)

Which statements are true regarding the Maximum Margin Hyperplane in SVM?

1. The classifier that contains a hyperplane with a small margin is more susceptible to model overfitting and tends to classify with weak confidence on unseen data.
2. The shortest distance from a hyperplane to one of its decision boundaries is equal to the shortest distance from the hyperplane to the decision boundary on its other side.

☐ 1 and 2 both are false
☐ 1 is true and 2 is false
☐ 1 is false and 2 is true
☐ 1 and 2 both are true

A developer is working on the SciPy library and importing the norm module by using the following code. Choose the correct output produced by this code.

```
from scipy.stats import norm  
after that, he writes the following code:  
print(norm.var())
```

☐ 1
☐ 0
☐ Compilation error
☐ Runtime error

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

You have multiple data frames, and you want to append these DataFrames without any information loss. Which of the following option do you use for this purpose?

☐ concat with axis=1 as a parameter
☐ concat with axis=0 as a parameter
☐ merge with axis=1 as a parameter
☐ join with axis=0 as a parameter

A survey on the average amount of time spent on smartphones was conducted. Typically, on the left, you see a list of all the things or activities people do with their phones, followed by a graph showing how much time they spend doing them. Which type of chart should be used for this purpose?

☒ matplotlib.pyplot.bar()
☐ plt.bar(x, height, width, bottom, at)
☐ matplotlib.pyplot.pie(data, explode, shadow=True)
☐ matplotlib.pyplot.scatter()
Clear Response

It has an employee dataset in which four columns are present, namely first name, last name, age, and salary. It wants to set the first name as an index column and store his dataset into a variable emp. Which of the following should he use for this purpose?

☐ emp.set_index("firstname", inplace=True)
☐ emp.set_index("firstname", inplace=False)
☐ emp.index="firstname"
☐ There is no such code

A developer is working on a NumPy array and writes the following code. Choose the correct output produced by this code.

```
import NumPy as np  
  
a= np.array([1, 2, 3, 4, 5, 6, 7,9,10,11,12,13,14])  
  
Print (a[-3:-2]).
```

☐ [12]
☐ [13]
☐ [11]
☐ [10]

Given is a DataFrame df = pd.DataFrame({"a": [1,4,5,2,6,9], "b": [0,3,2,1,8,7]})
What is the output of df.loc[[0,2,4]]?

☐ 10
52
68
☐ 12
01
68
☐ 12
52
79
☐ 12
42
86



While working on the Iris dataset, Raymond develops the following code using logistic regression. Choose the correct option to fill the missing lines in the code.

```
from sklearn.model_selection import train_test_split
from sklearn import datasets
#missing code 1

iris = datasets.load_iris() #Iris dataset

X_train, X_test, y_train, y_test = train_test_split(iris.data, iris.target,
                                                    test_size=0.3,
                                                    random_state=0)

g = #missing code 2
#missing code 3

fit(log_coef_)
```

- ☐ missing code 1= from sklearn.linear_model import LogisticRegression() missing code 2= LogisticRegression() missing code 3= log_fit(X_train,y_train)
- ☐ missing code 1= from sklearn import LogisticRegression() missing code 2= LogisticRegression() missing code 3= log_fit(X_train,y_train)
- ☐ missing code 1= from sklearn.linear_model import LogisticRegression() missing code 2= LogisticRegression() missing code 3= log_fit(y_train,X_train)
- ☐ missing code 1= from sklearn.linear_model import LogisticRegression() missing code 2= Regression() missing code 3= log_fit(X_train,y_train)

Suppose you need to evaluate regression model, how does the below functions help in evaluation?

1. cross_val_predict
2. Cross_val_score

- ☐ 1. Predicts the value of trained data for each set of features
2. Scores the value for any one splits of test
- ☐ 1. Predicts the value of tested data as average of all observations in tested split
2. Scores the value for any one splits of test
- ☐ 1. Predicts the value of test data for each set of features
2. Scores the value for different splits of test
- ☐ 1. Predicts the value of tested data as average of all observations in tested split
2. Scores the value for any different splits of test

Consider the given data set of population that contains the ages of users (x: 23,26,21,18,33,35,46,42,51,18). You are required to calculate the standard deviation of this set. Which of the given options represents the correct answer? Choose your answer from the listed choices.

- ☐ 11.98
- ☐ 11.36
- ☐ 12.55
- ☐ 10.65

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

Consider that a machine learning algorithm has given a very low accuracy than expected. This was performed on huge datasets where outliers or missing values are removed. Also extra information on dataset is extracted by feature transformation. How can you select the suitable features that are low dimensional space and will improve the accuracy of the algorithm?

- ☐ using PCA that reduces the dimensionality
- ☐ Using regression analysis
- ☐ Using PCA and dimensionality reduction by r
- ☐ using precision constant

Max, a machine learning developer, is working on a NumPy array. He has a NumPy array but he does not know the dimensions of his array. Which NumPy array attribute will help him to find out the dimensions of the array?

- ☐ ndim
- ☐ shape
- ☐ itemsize
- ☐ dtype

Consider that there is json file "students_details" with information of students as below

```
{
  "id": "1", "name": "student1"
}
{"id": "2", "name": "student2"}
{"id": "3", "name": "student3"}
{"id": "4", "name": "student4"}
}
```

How can you read json document with fields id and name?

- ☐ val dfs = sqlContext.read.json("student_details.json")
- ☐ val dfs = sqlContext.read.id("student_details.json")
- ☐ val dfs = sqlContext.read("student_details.json", "id", "name")
- ☐ val dfs = sqlContext.json("student_details")



Join our channel if you haven't joined yet https://t.me/fresco_milestone (@fresco_milestone)

Consider that a machine learning algorithm has given a very low accuracy than expected. This was performed on huge datasets where outliers or missing values are removed. Also extra information on dataset is extracted by feature transformation. How can you select the suitable features that are low dimensional space and will improve the accuracy of the algorithm?

A developer is working on data preprocessing and wants to scale his dataset between a given minimum and maximum value. Which of the following scikit-learn modules should he use for this?

- ☐ using PCA that reduces the dimensionality
- ☐ Using regression analysis
- ☐ Using PCA and dimensionality reduction by regression analysis
- ☐ using precision constant
- ☐ sklearn.preprocessing.StandardScaler()
- ☐ sklearn.preprocessing.MinMaxScaler()
- ☐ sklearn.preprocessing.QuantileTransformer()
- ☐ There is no such module

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

Tony is working on the following NumPy array:

```
age=np.array([[22,25,27,30,32,34,37,38,40,50]])
```

He performs some array manipulation operations on the above array and gets the following output. Which code gives the same output as below?

```
[[22]
 [25]
 [27]
 [30]
 [32]
 [34]
 [37]
 [38]
 [40]
 [50]]
```

- ☐ import numpy as np

age=np.array([[22,25,27,30,32,34,37,38,40,50]])
print(age.T)
- ☐ import numpy as np

age=np.array([22,25,27,30,32,34,37,38,40,50])
print(age.T)
- ☐ import numpy as np

age=np.array([22,25,27,30,32,34,37,38,40,50])
print(age.I)
- ☐ "import numpy as np

age=np.array([[22,25,27,30,32,34,37,38,40,50]])
print(age.I)



Consider that there is json file "students_details" with information of students as below

```
{
  "id": "1", "name": "student1"
}
{
  "id": "2", "name": "student2"
}
{
  "id": "3", "name": "student3"
}
{
  "id": "4", "name": "student4"
}
```

How can you read json document with fields id and name?

- ☐ `val dfs = sqlContext.read.json("student_details.json")`
- ☐ `val dfs = sqlContext.read.id("student_details.json")`
- ☐ `val dfs = sqlContext.read("student_details.json", "id", "name")`
- ☐ `val dfs = sqlContext.json("student_details")`

While working on a movie review dataset, we take three reviews and perform a bag of words. What are the vectors for review1, review2, and review3?

- Review 1: This movie is very scary and long
- Review 2: This movie is not scary and is slow
- Review 3: This movie is spooky and good

- ☐ Vector of Review 1: [1 1 1 1 1 1 0 0 0 0]
Vector of Review 2: [1 1 2 0 0 1 1 0 1 0]
Vector of Review 3: [1 1 1 0 0 0 1 0 0 1 1]
- ☐ *Vector of Review 1: [1 0 1 0 0 0 1 0 0 1 0]
Vector of Review 2: [1 1 3 0 0 1 1 0 1 0]
Vector of Review 3: [1 1 0 1 0 1 1 1 0 0 0]
- ☐ Vector of Review 1: [1 1 1 1 1 1 1 0 0 0]
Vector of Review 2: [1 1 1 0 0 0 1 0 0 1 0]
Vector of Review 3: [1 1 2 0 0 1 1 0 1 0 0]
- ☐ Vector of Review 1: [1 0 1 0 0 0 1 0 0 1 0]
Vector of Review 2: [1 1 0 1 0 1 1 1 0 0 0]
Vector of Review 3: [1 1 3 0 0 1 1 0 1 0 0]

What will be the regular expression for a binary string that represents an even number?

e.g. Strings like the below should be accepted:

10
1110
11000110110

While strings like the below should be rejected:

010
00001110
1
101
1101

- ☐ $((0+1)^*0)^*$
- ☐ $(0+1)^*0$
- ☐ $(0+1)(1)^*$
- ☐ $^*1((0+1)^*0)$

While evaluating the machine learning algorithm performance on supervised learning, you need to split the dataset in to two sets. one for fitting in to the ML model called trainset and another evaluating the same model called testset. So which of the following options performs this above mentioned task?

- ☐ `X_train, X_test, Y_train, Y_test`
- ☐ `X_train, X_test, Y_train, Y_test`
- ☐ `X_train, Y_test= train_test_sp`
- ☐ `X_train, Y_test, = train_split(`



@FRESCO_MILESTONE
CLICK HERE



A developer stores the following into a variable n:
[[2.5, 2.4], [1.5, 0.7], [5.2, 2.9], [19, 1.2], [1.1, 3.0], [3.1, 1.9]]

He wants to find the principal component of the above array. Which function should he use for this?

- ☐ from sklearn.decomposition import PCA
import numpy as np

n = np.array([[2.5, 2.4], [1.5, 0.7], [5.2, 2.9], [19, 1.2], [1.1, 3.0], [3.1, 1.9]])
pca = PCA()
pca.fit(n)

eigenvectors
print(pca.components_)
- ☐ from sklearn.model_selection import PCA
import numpy as np

n = np.array([[2.5, 2.4], [1.5, 0.7], [5.2, 2.9], [19, 1.2], [1.1, 3.0], [3.1, 1.9]])
pca = PCA()
pca.fit(n)

eigenvectors
print(pca.components_)
- ☐ from sklearn.decomposition import PCA
import numpy as np

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

Consider the given data set of population that contains the ages of users (i.e. 23, 28, 21, 18, 33, 35, 45, 42, 51, 18). You are required to calculate the standard deviation of this set. Which of the given options represents the correct answer? Choose your answer from the listed choices.

☒ 11.98
☐ 11.36
☐ 12.55
☐ 10.55

Clear Response

A developer wants to preprocess his data using standard scaling, and he stores this data into a variable such as:
data = [[0, 0], [0, 5], [1, 2], [1, 5]]

He wants the output to be in the form:
[[-1. -0.96225045]
[-1. -0.70565033]
[1. 0.06415003]
[1. 1.60375075]]

Which of the following code shall he use for obtaining the above output?

- ☐ from sklearn.preprocessing import StandardScaler
data = [[0, 0], [0, 5], [1, 2], [1, 5]]
scaler = StandardScaler()
scaler.fit(data)
print(scaler.transform(data))
- ☐ from sklearn.preprocessing import StandardScaler
data = [[0, 0], [0, 5], [1, 2], [1, 5]]
scaler = StandardScaler()
print(scaler.transform(data))
- ☐ from sklearn.preprocessing import StandardScaler
data = [[0, 0], [0, 5], [1, 2], [1, 5]]
scaler = StandardScaler()
scaler.fit(data)
print(scaler.mean(data))
- ☐ from sklearn.preprocessing import StandardScaler
data = [[0, 0], [0, 5], [1, 2], [1, 5]]
scaler = StandardScaler()

print(scaler.mean(data))

What will be the regular expression for a binary string that represents an even number?

e.g. Strings like the below should be accepted:
10
1110
110001101110

While strings like the below should be rejected:
010
00001110
1
101
1101

- ☐ ((0+1)*0)*
- ☐ (0+1)*0
- ☐ (0+1)(1)*
- ☐ *1((0+1)*0)

Note: This file is posted in our channel for free. (https://t.me/fresco_milestone).

